

Introduction

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There is a general awareness that contemporary physics has brought about an important revision in man's conception of the universe and his relation to it. The suggestion has been made that this revision pierces to the basis of man's fate and freedom, affecting even his conception of his capacity to control his own destiny. In no portion of physics does this suggestion show itself more pointedly than in the principle of indeterminacy of quantum mechanics. The author of this book is the discoverer of this principle. In fact, it usually bears his name. Hence, no one is more competent to pass judgment on what it means than he.

In his previous book, *The Physical Principles of the Quantum Theory*,* Heisenberg gave an exposition of the theoretical interpretation, experimental meaning and mathematical apparatus of quantum mechanics for professional physicists. Here he pursues this and other physical theories with respect to their philosophical implications and some of their likely social consequences for the layman. More specifically, he attempts here to raise and suggest answers to three questions: (1) What do the experimentally verified theories of contemporary physics affirm? (2) How do they permit or require man to think of himself in relation to his universe? (3) How is this new way of

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thinking, which is the creation of the modern West, going to affect other parts of the world?

The third of these questions is dealt with briefly by Heisenberg at the beginning and end of this inquiry. The brevity of his remarks should not lead the reader to pass lightly over their import. As he notes, whether we like it or not, modern ways are going to alter and in part destroy traditional customs and values. It is frequently assumed by native leaders of non-Western societies, and also often by their Western advisers, that the problem of introducing modern scientific instruments and ways into Asia, the Middle East and Africa is merely that of giving the native people their political independence and then providing them with the funds and the practical instruments. This facile assumption overlooks several things. First, the instruments of modern science derive from its theory and require a comprehension of that theory for their correct manufacture or effective use. Second, this theory in turn rests on philosophical, as well as physical, assumptions. When comprehended, these philosophical assumptions generate a personal and social mentality and behavior quite different from, and at points incompatible with, the family, caste and tribally centered mentality and values of the native Asian, Middle Eastern or African people. In short, one cannot bring in the instruments of modern physics without sooner or later introducing its philosophical mentality, and this mentality, as it captures the scientifically trained youth, upsets the old familial and tribal moral loyalties. If unnecessary emotional conflict and social demoralization are not to result, it is important that the youth understand what is happening to them. This means that they must see their experience as the coming together of two different philosophical mentalities, that of their traditional culture and that of the new

physics. Hence, the importance for everyone of understanding the philosophy of the new physics.

But it may be asked, Isn't physics quite independent of philosophy? Hasn't modern physics become effective only by dropping philosophy? Clearly, Heisenberg answers both of these questions in the negative. Why is this the case?

Newton left the impression that there were no assumptions in his physics which were not necessitated by the experimental data. This occurred when he suggested that he made no hypotheses and that he had deduced his basic concepts and laws from the experimental findings. Were this conception of the relation between the physicist's experimental observations and his theory correct, Newton's theory would never have required modification, nor could it ever have implied consequences which experiment does not confirm. Being implied by the facts, it would be as indubitable and final as they are.

In 1885, however, an experiment performed by Michelson and Morley revealed a fact which should not exist were the theoretical assumptions of Newton the whole truth. This made it evident that the relation between the physicist's experimental facts and his theoretical assumptions is quite other than what Newton had led many modern physicists to suppose. When, some ten years later, experiments on radiation from black bodies enforced an additional reconstruction in Newton's way of thinking about his subject matter, this conclusion became inescapable. Expressed positively, this means that the theory of physics is neither a mere description of experimental facts nor something deducible from such a description; instead, as Einstein has emphasized, the physical scientist only arrives at his theory by speculative means. The deduction in his method runs not from facts to the assumptions of the theory but from the assumed

theory to the facts and the experimental data. Consequently, theories have to be proposed speculatively and pursued deductively with respect to their many consequences so that they can be put to indirect experimental tests. In short, any theory of physics makes more physical and philosophical assumptions than the facts alone give or imply. For this reason, any theory is subject to further modification and reconstruction with the advent of new evidence that is incompatible, after the manner of the results of the Michelson-Morley experiment, with its basic assumptions.

These assumptions, moreover, are philosophical in character. They may be ontological, i.e., referring to the subject matter of scientific knowledge which is independent of its relation to the perceiver; or they may be epistemological, i.e., referring to the relation of the scientist as experimenter and knower to the subject matter which he knows. Einstein's special and general theories of relativity modify the philosophy of modern physics in the first of these two respects by radically altering the philosophical theory of space and time and their relation to matter. Quantum mechanics, especially its Heisenberg principle of indeterminacy, has been notable for the change it has brought in the physicist's epistemological theory of the relation of the experimenter to the object of his scientific knowledge. Perhaps the most novel and important thesis of this book is its author's contention that quantum mechanics has brought the concept of potentiality back into physical science. This makes quantum theory as important for ontology as for epistemology. At this point, Heisenberg's philosophy of physics has an element in common with that of Whitehead.

It is because of this introduction of potentiality into the subject matter of physics, as distinct from the epistemological

predicament of physicists, that Einstein objected to quantum mechanics. He expressed this objection by saying: "God does not play dice." The point of this statement is that the game of dice rests on the laws of chance, and Einstein believed that the latter concept finds its scientific meaning solely in the epistemological limitations of the finite knowing mind in its relation to the omnicomplete object of scientific knowledge and, hence, is misapplied when referred ontologically to that object itself. The object being per se all complete and in this sense omniscient, after the manner of God, the concept of chance or of probability is inappropriate for any scientific description of it.

This book is important because it contains Heisenberg's answer to this criticism of his principle of indeterminacy and of quantum theory by Einstein and by others. In understanding this answer two things must be kept in mind: (1) The aforementioned relation between the data of experimental physics and the concepts of its theory. (2) The difference between the role of the concept of probability in (a) Newton's mechanics and Einstein's theory of relativity and in (b) quantum mechanics. Upon (1), Einstein and Heisenberg, and relativistic mechanics and quantum mechanics, are in agreement. It is only with respect to (2) that they differ. Yet the reason for Heisenberg's and the quantum physicist's difference from Einstein on (2) depends in considerable part on (1) which Einstein admits.

(1) affirms that the experimental data of physics do not imply its theoretical concepts. From this it follows that the object of scientific knowledge is never known directly by observation or experimentation, but is only known by speculatively proposed theoretic construction or axiomatic postulation, tested only indirectly and experimentally via its deduced consequences. To

find the object of scientific knowledge we must go, therefore, to its theoretical assumptions.

When we do this for (a) Newton's or Einstein's mechanics and for (b) quantum mechanics, we discover that the concept of probability or chance enters into the definition of the state of a physical system, and, in this sense, into its subject matter, in quantum mechanics, but does not do so in Newton's mechanics or Einstein's theory of relativity. This undoubtedly is what Heisenberg means when he writes in this book that quantum theory has brought the concept of potentiality back into physical science. It is also, without question, what Einstein has in mind when he objects to quantum theory.

Put more concretely, this difference between quantum mechanics and previous physical theories may be expressed as follows: In Newton's and Einstein's theory, the state of any isolated mechanical system at a given moment of time is given precisely when only numbers specifying the position and momentum of each mass in the system are empirically determined at that moment of time; no numbers referring to a probability are present. In quantum mechanics the interpretation of an observation of a system is a rather complicated procedure. The observation may consist in a single reading, the accuracy of which has to be discussed, or it may comprise a complicated set of data, such as the photograph of the water droplets in a cloud chamber; in any case, the result can be stated only in terms of a probability distribution concerning, for instance, the position or momentum of the particles of the system. The theory then predicts the probability distribution for a future time. The theory is not experimentally verified when that future state arrives if merely the momentum or position numbers in a particular observation lie within the predicted range. The same experiment

with the same initial conditions must be repeated many times, and the values of position or momentum, which may be different in each observation, must similarly be found to be distributed according to the predicted probability distribution. In short, the crucial difference between quantum mechanics and Einstein's or Newton's mechanics centers in the definition of a mechanical system at any moment of time, and this difference is that quantum mechanics introduces the concept of probability into its definition of state and the mechanics of Newton and Einstein does not.

This does not mean that probability had no place in Newton's or Einstein's mechanics. Its place was, however, solely in the theory of errors by means of which the accuracy of the Yes or No verification or nonconfirmation of the prediction of the theory was determined. Hence, the concept of probability and chance was restricted to the epistemological relation of the scientist in the verification of what he knows; it did not enter into the theoretical statement of what he knows. Thus, Einstein's dictum that "God does not play dice" was satisfied in his two theories of relativity and in Newton's mechanics.

Is there any way of deciding between Einstein's contention and that of Heisenberg and other quantum theorists? Many answers have been given to this question. Some physicists and philosophers, emphasizing operational definitions, have argued that, since all physical theories, even classical ones, entail human error and uncertainties, there is nothing to be decided between Einstein and the quantum theorists. This, however, is (a) to overlook the presence of axiomatically constructed, constitutive theoretic definitions as well as theory-of-errors, operational definitions in scientific method and (b) to suppose that the concept of probability and the even more complex uncertainty

relation enter into quantum mechanics only in the operational-definitional sense. Heisenberg shows that the latter supposition is false.

Other scientists and philosophers, going to the opposite extreme, have argued that, merely because there is uncertainty in predicting certain phenomena, this constitutes no argument whatever for the thesis that these phenomena are not completely determined. This argument combines the statical problem of defining the state of a mechanical system at a given time with the dynamical or causal problem of predicting changes in the state of the system through time. But the concept of probability in quantum theory enters only into its statics, i.e., its theoretical definition of state. The reader will find it wise, therefore, to keep distinct these two components, i.e., the statical theoretical definition-of-state component and the dynamic, or causal, theoretical change-of-state-through-time component. With respect to the former, the concept of probability and the attendant uncertainty enter theoretically and in principle; they do not refer merely to the operational and epistemological uncertainties and errors, arising from the finiteness of, and inaccuracies in, human behavior, that are common to any scientific theory and any experimentation whatsoever.

But, why, it may be asked, should the concept of probability be introduced into the theoretic definition of the state of a mechanical system at any statical moment t^1 *in principle*? In making such a theoretical construct by axiomatic postulation, do not Heisenberg and quantum theoreticians generally beg the question at issue between themselves and Einstein? This book makes it clear that the answer to these questions is as follows: The reason for the procedure of quantum mechanics is thesis (1) above, which Einstein himself also accepts.

Thesis (1) is that we know the object of scientific knowledge only by the speculative means of axiomatic theoretic construction or postulation; Newton's suggestion that the physicist can deduce our theoretical concepts from the experimental data being false. It follows that there is no *a priori* or empirical meaning for affirming that the object of scientific knowledge, or, more specifically, the state of a mechanical system at a given time t^1 , must be defined in a particular way. The sole criterion is, which set of theoretic assumptions concerning the subject matter of mechanics when pursued to their deduced experimental consequences is confirmed by the experimental data?

Now, it happens that when we theoretically and in principle define the state of a mechanical system for subatomic phenomena in terms solely of numbers referring to position and momentum, as Einstein would have us do, and deduce the consequences for radiation from black bodies, this theoretical assumption concerning the state of a mechanical system and the subject matter of atomic physics is shown to be false by experimental evidence. The experimental facts simply are not what the theory calls for. When, however, the traditional theory is modified with the introduction of Planck's constant and the addition in principle of the second set of numbers referring to the probability that the attached position-momentum numbers will be found, from which the uncertainty principle follows, the experimental data confirm the new theoretical concepts and principles. In short, the situation in quantum mechanics with respect to experiments on black-body radiation is identical with that faced by Einstein with respect to the Michelson-Morley experiment. In both cases, only by introducing the new theoretical assumption in principle is physical theory brought into accord with the experimental facts. Thus, to assert that, notwithstanding quantum

mechanics, the positions and momenta of subatomic masses are “really” sharply located in space and time as designated by one pair of numbers only and, hence, completely deterministic causally, as Einstein and the aforementioned philosophers of science would have one do, is to affirm a theory concerning the subject matter of physical knowledge which experiments on black-body radiation have shown to be false in the sense that a deductive experimental consequence of this theory is not confirmed.

It does not follow, of course, that some new theory compatible with the foregoing experimental facts might not be discovered in which the concept of probability does not enter in principle into its definition of state. Professor Norbert Wiener, for example, believes that he has clues to the direction such a theory might take. It would, however, have to reject a definition of state in terms of the four space-time dimensions of Einstein’s theory and would, therefore, be incompatible with Einstein’s thesis on other grounds. Certainly, one cannot rule out such a possibility. Nevertheless, until such an alternative theory is presented, anyone, who does not claim to possess some *a priori* or private source of information concerning what the object of scientific knowledge must be, has no alternative but to accept the definition of state of quantum theory and to affirm with the author of this book that it restores the concept of potentiality to the object of modern scientific knowledge. Experiments on black-body radiation require one to conclude that God plays dice.

What of the status of causality and determinism in quantum mechanics? Probably the interest of the layman and the humanist in this book depends most on its answer to this question.

If this answer is to be understood, the reader must pay particular attention to Heisenberg’s description of (a) the aforementioned definition of state by recourse to the concept of

probability and (b) the Schrödinger time-equation. The reader must also make sure, and this is the most difficult task of all, that the meaning of the words "causality" and "determinism" in his mind when he asks the above question is identical with the meaning these words have in Heisenberg's mind when he specifies the answer. Otherwise, Heisenberg will be answering a different question from the one the reader is asking and a complete misunderstanding upon the reader's part will occur.

The situation is further complicated by the fact that modern physics permits the concept of causality to have two different, scientifically precise meanings, the one stronger than the other, and there is no agreement among physicists about which one of these two meanings the word "causality" is to be used to designate. Hence, some physicists and philosophers of science use the word to designate the stronger of the two meanings. There is evidence, at times at least, that this is Professor Heisenberg's usage in this book. Other physicists and philosophers, including the writer of this Introduction, use the word "causality" to designate the weaker of the two meanings and the word "determinism" to designate the stronger meaning. When the former usage is followed, the words "causality" and "determinism" become synonymous. When the second usage is followed, every deterministic system is a causal system, but not every causal system is deterministic.

Great confusion has entered into previous discussion of this topic because frequently neither the person who asks the question nor the physicist who has answered it has been careful to specify in either question or answer whether he is using the word "causality" in its weaker or in its stronger modern scientific meaning. If one asks "Does causality hold in quantum mechanics?" not specifying whether one is asking about causality

in its stronger or in its weaker sense, one then gets apparently contradictory answers from equally competent physicists. One physicist, taking the word "causality" in its stronger sense, quite correctly answers "No." The other physicist, taking "causality" in its weaker sense, equally correctly answers "Yes." Naturally, the impression has arisen that quantum mechanics is not specific about what the answer is. Nevertheless, this impression is erroneous. The answer of quantum mechanics becomes unequivocal the moment one makes the question and the answer unambiguous by specifying which meaning of "causality" one is talking about.

It is important, therefore, to become clear about different possible meanings of the word "causality." Let us begin with the layman's common-sense usage of the word "cause" and then move to the more exact meanings in modern physics, considering the meaning in Aristotle's physics on the way.

One may say "The stone hit the window and *caused* the glass to break." In this use of "causality" it is thought of as a relation between objects, i.e., between the stone and the windowpane. The scientist expresses the same thing in a different way. He describes the foregoing set of events in terms of the state of the stone and the windowpane at the earlier time t^1 when the stone and the windowpane were separated and the state of this same system of two objects at the later time t^2 when the stone and the windowpane collided. Consequently, whereas the layman tends to think of causality as a relation between objects, the scientist thinks of it as a relation between different states of the same object or the same system of objects at different times.

This is why, in order to determine what quantum mechanics says about causality, one must pay attention to two things: (1) The state-function which defines the state of any physical system

at any specific time t . (2) The Schrödinger time-equation which relates the state of the physical system at the earlier time t^1 to its different state at any specifiable later time t^2 . What Heisenberg says about (1) and (2) must, therefore, be read with meticulous care.

It will help to understand what quantum mechanics says about the relation between the states of a given physical object, or system of physical objects, at different times if we consider the possible properties that this relation might have. The weakest possible case would be that of mere temporal succession with no necessary connection whatever and with not even a probability, however small, that the specifiable initial state will be followed in time by a specifiable future state. Hume gives us reasons for believing that the relation between the sensed states of immediately sensed natural phenomena is of this character. Certainly, as he pointed out, one does not sense any relation of necessary connection. Nor does one directly sense probability. All that sensation gives us with respect to the successive states of any phenomenon is the mere relation of temporal succession.

This point is of great importance. It means that one can arrive at a causal theory in any science or in common-sense knowledge, or even at a probability theory, of the relation between the successive states of any object or system, only by speculative means and axiomatically constructed, deductively formulated scientific and philosophical theory which is tested not directly against the sensed and experimental data but only indirectly by way of its deductive consequences.

A second possibility with respect to the character of the relation between the states of any physical system at different times is that the relation is a necessary one, but that one can know what this necessary connection is only by knowing the future

state. The latter knowledge of the future state may be obtained either by waiting until it arrives or by having seen the future or final state of similar systems in the past. When such is the case, causality is teleological. Changes of the system with time are determined by the final state or goal of the system. The physical system which is an acorn in the earlier state t^1 and an oak tree in the later state t^2 is an example. The connection between these two states seems to be a necessary one. Acorns never change into maple trees or into elephants. They change only into oaks. Yet, given the properties of this physical system in the acorn state of the earlier time t^1 , no scientist has as yet been able to deduce the properties of the oak tree which the system will have at the later time t^2 . Aristotelian physics affirmed that all causal relations are teleological.

Another possibility is that the relation between the states of any object, or any system of objects, at different times is a relation of necessary connection such that, given knowledge of the initial state of the system, assuming isolation, its future state can be deduced. Stated in more technical mathematical language, this means that there exists an indirectly verified, axiomatically constructed theory whose postulates (1) specify a state-function, the independent variables of which completely define the state of the system at any specific instant of time, and (2) provide a time-equation relating the numerical empirical values of the independent variables of this function at any earlier time t^1 to their numerical empirical values at any specific later time t^2 in such a way that by introducing the operationally determined t^1 set of numbers into the time-equation the future t^2 numbers can be deduced by merely solving the equation. When this is the case, the temporal relation between states is said to exemplify mechanical causation.

It is to be noted that this definition of mechanical causality leaves open the question of what independent variables are required to define the state of the system at any given time. Hence, at least two possibilities arise: (a) the concept of probability may be used to define the state of the system or (b) it may not be so used. When (b) is the case no independent variables referring to probabilities appear in the state-function and the stronger type of mechanical causality is present. When (a) is the case independent variables referring to probabilities, as well as to other properties such as position and momentum, appear in the state-function and only the weaker type of mechanical causation occurs. If the reader keeps these two meanings of mechanical causation in mind and makes sure which meaning Heisenberg is referring to in any particular sentence of this book, he should be able to get its answer to the question concerning the status of causality in modern physics.

What of determinism? Again, there is no agreed-upon convention among physicists and philosophers of science about how this word is to be used. It is in accord with the common-sense usage to identify it with the strongest possible causality. Let us, then, use the word "determinism" to denote only the stronger type of mechanical causation. Then I believe the careful reader of this book will get the following answer to his question: In Newtonian, Einsteinian and quantum mechanics, mechanical, rather than teleological, causality holds. This is why quantum physics is called quantum mechanics, rather than quantum teleologics. But, whereas causality in Newton's and Einstein's physics is of the stronger type and, hence, both mechanical and deterministic, in quantum mechanics it is of the weaker causal type and, hence, mechanical but not deterministic. From the latter fact it follows that if anywhere in this book Heisenberg

uses the words "mechanical causality" in their stronger, deterministic meaning and the question be asked "Does mechanical causation in this stronger meaning hold in quantum mechanics?" then the answer has to be "No."

The situation is even more complicated, as the reader will find, than even these introductory distinctions between the different types of causation indicate. It is to be hoped, however, that this focusing of attention upon these different meanings will enable the reader to find his way through this exceptionally important book more easily than would otherwise be the case.

These distinctions should suffice, also, to enable one to grasp the tremendous philosophical significance of the introduction of the weaker type of mechanical causation into modern physics, which has occurred in quantum mechanics. Its significance consists in reconciling the concept of objective, and in this sense ontological, potentiality of Aristotelian physics with the concept of mechanical causation of modern physics.

It would be an error, therefore, if the reader, from Heisenberg's emphasis upon the presence in quantum mechanics of something analogous to Aristotle's concept of potentiality, concluded that contemporary physics has taken us back to Aristotle's physics and ontology. It would be an equal error conversely to conclude, because mechanical causation in its weaker meaning still holds in quantum mechanics, that all is the same now in modern physics with respect to its causality and ontology as was the case before quantum mechanics came into being. What has occurred is that in quantum theory contemporary man has moved on beyond the classical medieval and the modern world to a new physics and philosophy which combines consistently some of the basic causal and ontological assumptions of each. Here, let it be recalled, we use the word "ontological"

to denote any experimentally verified concept of scientific theory which refers to the object of scientific knowledge rather than merely to the epistemological relation of the scientist as knower to the object which he knows. Such an experimentally verified philosophical synthesis of ontological potentiality with ontological mechanical causality, in the weaker meaning of the latter concept, occurred when physicists found it impossible to account theoretically for the Compton effect and the results of experiment on black-body radiation unless they extended the concept of probability from its Newtonian and Einsteinian merely epistemological, theory-of-errors role in specifying when their theory is or is not experimentally confirmed to the ontological role, specified in principle in the theory's postulates, of characterizing the object of scientific knowledge itself.

Need one wonder that Heisenberg went through the subjective emotional experiences described in this book before he became reconciled to the necessity, imposed by both experimental and mathematical considerations, of modifying the philosophical and scientific beliefs of both medieval and modern man in so deep-going a manner. Those interested in a firsthand description of the human spirit in one of its most creative moments will want to read this book because of this factor alone. The courage which it took to make this step away from the unqualified determinism of classical modern physics may be appreciated if one recalls that even such a daring, creative spirit as Einstein balked. He could not allow God to play dice; there could not be potentiality in the object of scientific knowledge, as the weaker form of mechanical causality in quantum mechanics allows.

Before one concludes, however, that God has become a complete gambler and that potentiality is in all objects, certain limi-

tations which quantum mechanics places on the application of its weaker form of mechanical causation must be noted. To appreciate these qualifications the reader must note what this book says about (1) the Compton effect, (2) Planck's constant h , and (3) the uncertainty principle which is defined in terms of Planck's constant.

This constant h is a number referring to the quantum of action of any object or system of objects. This quantum, which extends atomicity from matter and electricity to light and even to energy itself, is very small. When the quantum numbers of the system being observed are small, as is the case with subatomic phenomena, then the uncertainty specified by the Heisenberg uncertainty principle of the positions and momenta of the masses of the system becomes significant. Then, also, the probability numbers associated with the position-momentum numbers in the state-function become significant. When, however, the quantum numbers of the system are large, then the quantitative amount of uncertainty specified by the Heisenberg principle becomes insignificant and the probability numbers in the state-function can be neglected. Such is the case with gross common-sense objects. At this point quantum mechanics with its basically weaker type of causality gives rise, as a special case of itself, to Newtonian and Einsteinian mechanics with their stronger type of causality and determinism. Consequently, for human beings considered *merely* as gross common-sense objects the stronger type of causality holds and, hence, determinism reigns also.

Nevertheless, subatomic phenomena are scientifically significant in man. To this extent, at least, the causality governing him is of the weaker type, and he embodies both mechanical fate and potentiality. There are scientific reasons for believing that this occurs even in heredity. Any reader who wants to pursue this

topic beyond the pages of this book should turn to *What Is Life?** by Professor Erwin Schrödinger, the physicist after whom the time-equation in quantum mechanics is named. Undoubtedly, potentiality and the weaker form of causality hold also for countless other characteristics of human beings, particularly for those cortical neural phenomena in man that are the epistemic correlates of directly introspected human ideas and purposes.

If the latter possibility is the case, the solution of a baffling scientific, philosophical and even moral problem may be at hand. This problem is: How is the mechanical causation, even in its weaker form, of quantum mechanics to be reconciled with the teleological causation patently present in the moral, political and legal purposes of man and in the teleological causal determination of his bodily behavior, in part at least, by these purposes? In short, how is the philosophy of physics expounded in this book by Heisenberg to be reconciled with moral, political and legal science and philosophy?

It may help the reader to appreciate why this book must be mastered before these larger questions can be correctly understood or effectively answered if very brief reference is made here to some articles which relate its theory of physical causation to the wider relation between mechanism and teleology in the humanities and the social sciences. The relevant articles are (a) by Professors Rosenblueth, Wiener and Bigelow in the journal of *The Philosophy of Science* for January, 1943; (b) by Doctors McCulloch and Pitts in *The Bulletin of Mathematical Biophysics*, Volume 5, 1943, and Volume 9, 1947; and (c) Chapter XIX of *Ideological Differences and World Order*, edited by the writer of this Introduction and published by the Yale University

* University Press, Cambridge; Macmillan Company, New York; 1946.

Press in 1949. If read after this book, (a) will show how teleological causality arises as a special case of the mechanical causality described by Heisenberg here. Similarly, (b) will provide a physical theory of the neurological correlates of introspected ideas, expressed in terms of the teleologically mechanical causality of (a), thereby giving an explanation of how ideas can have a causally significant effect on the behavior of men. Likewise, (c) will show how the ideas and purposes of moral, political and legal man relate, by way of (b) and (a), to the theory of physical potentiality and mechanical causality so thoroughly described by Heisenberg in this book.

It remains to call attention to what Professor Heisenberg says about Bohr's principle of complementarity. This principle plays a great role in the interpretation of quantum theory by "the Copenhagen School" to which Bohr and Heisenberg belong. Some students of quantum mechanics, such as Margenau in his book *The Nature of Physical Reality*,* are inclined to the conclusion that quantum mechanics requires merely its definition of state, its Schrödinger time-equation and those other of its mathematical postulates which suffice to ensure, as noted above, that Einsteinian and Newtonian mechanics come out of quantum mechanics as one of its special cases. According to the latter thesis, the principle of complementarity arises from the failure to keep the stronger and weaker form of mechanical causality continuously in mind, with the resultant attribution of the stronger form to those portions of quantum mechanics where only the weaker form is involved. When this happens, the principle of complementarity has to be introduced to avoid

* McGraw-Hill Book Co., Inc., New York, 1950, pp. 418-422. See also, Northrop, *The Logic of the Sciences and the Humanities*, Macmillan, New York, 1947, Chapter XI.

contradiction. If, however, one avoids the foregoing practice, the principle of complementarity becomes, if not unnecessary, at least of a form such that one avoids the danger, noted by Margenau * and appreciated by Bohr, of giving pseudo solutions to physical and philosophical problems by playing fast and loose with the law of contradiction, in the name of the principle of complementarity.

By its use the qualifications that had to be put on both the particle-picture common-sense language of atomic physics and its common-sense wave-picture language were brought together. But once having formulated the result with axiomatically constructed mathematical exactitude, any further use of it is merely a superficial convenience when, leaving aside the exact and essential mathematical assumptions of quantum mechanics, one indulges in the common-sense language and images of waves and particles.

It has been necessary to go into the different interpretations of the principle of complementarity in order to enable the reader to pass an informed judgment concerning what Heisenberg says in this book about the common-sense and Cartesian concepts of material and mental substances. This is the case because his conclusion concerning Descartes results from his generalization of the principle of complementarity beyond physics, first, to the relation between common-sense biological concepts and mathematical physical concepts and, second, to the body-mind problem. The result of this generalization is that the Cartesian theory of mental substances comes off very much better in this book, as does the concept of substance generally, than is the case in any other book on the philosophy of contemporary physics which this writer knows.

* Margenau, *op. cit.*, p. 422.

Whitehead, for example, concluded that contemporary science and philosophy find no place for, and have no need of, the concept of substance. Neutral monists such as Lord Russell and logical positivists such as Professor Carnap agree.

Generally speaking, Heisenberg argues that there is no compelling reason to throw away any of the common-sense concepts of either biology or mathematical physics, after one knows the refined concepts that lead to the complete clarification of the problems in atomic physics. Because the latter clarification is complete, it is relevant only to a very limited range of problems within science and cannot enable us to avoid using many concepts at other places that would not stand critical analysis of the type carried out in quantum theory. Since the ideal of complete clarification cannot be achieved—and it is important that we should not be deceived about this point—one may indulge in the usage of common-sense concepts if it is done with sufficient care and caution. In this respect, certainly, complementarity is a very useful scientific concept.

In any event, two things seem clear and make what Heisenberg says on these matters exceedingly important. First, the principle of complementarity and the present validity of the Cartesian and common-sense concepts of body and mind stand and fall together. Second, it may be that both these notions are merely convenient stepladders which should now be, or must eventually be, thrown away. Even so, in the case of the theory of mind at least, the stepladder will have to remain until by its use we find the more linguistically exact and empirically satisfactory theory that will permit us to throw the Cartesian language away. To be sure, piecemeal theories of mind which do not appeal to the notion of substance now exist, but none of their authors, unless it be Whitehead, has shown how the lan-

gauge of this piecemeal theory can be brought into commensurate and compatible relationship with the scientific language of the other facts of human knowledge. It is likely, therefore, that anyone, whether he be a professional physicist or philosopher or the lay reader, who may think he knows better than Heisenberg on these important matters, runs the grave risk of supposing he has a scientific theory of mind in its relation to body, when in fact this is not the case.

Up to this point we have directed attention, with but two exceptions, to what the philosophy of contemporary physics has to say about the object of scientific knowledge *qua* object, independent of its relation to the scientist as knower. In short, we have been concerned with its ontology. This philosophy also has its epistemological component. This component falls into three parts: (1) The relation between (a) the directly observed data given to the physicist as inductive knower in his observations or his experiments and (b) the speculatively proposed, indirectly verified, axiomatically constructed postulates of his theory. The latter term (b) defines the objects of scientific knowledge *qua* object and, hence, gives the ontology. The relation between (a) and (b) defines one factor in the epistemology. (2) The role of the concept of probability in the theory of errors, by means of which the physicist defines the criterion for judging how far his experimental findings can depart, due to errors of human experimentation, from the deduced consequences of the postulates of the theory and still be regarded as confirming the theory. (3) The effect of the experiment being performed upon the object being known. What Heisenberg says about the first and second of these three epistemological factors in contemporary physics has already been emphasized in this Introduction. It remains to direct the reader's attention to what he says about item (3).

In modern physical theory, previous to quantum mechanics, (3) played no role whatever. Hence, the epistemology of modern physics was then completely specified by (1) and (2) alone. In quantum mechanics, however, (3) (as well as (1) and (2)) becomes very important. The very act of observing alters the object being observed when its quantum numbers are small.

From this fact Heisenberg draws a very important conclusion concerning the relation between the object, the observing physicist, and the rest of the universe. This conclusion can be appreciated if attention is directed to the following key points. It may be recalled that in some of the definitions of mechanical causality given earlier in this Introduction, the qualifying words "for an isolated system" were added; elsewhere it was implicit. This qualifying condition can be satisfied in principle in Newtonian and Einsteinian mechanics, and also in practice by making more and more careful observations and refinements in one's experimental instruments. The introduction of the concept of probability into the definition of state of the object of scientific knowledge in quantum mechanics rules out, however, *in principle*, and not merely in practice due to the imperfections of human observation and instruments, the satisfying of the condition that the object of the physicist's knowledge is an isolated system. Heisenberg shows also that the including of the experimental apparatus and even of the eye of the observing scientist in the physical system which is the object of the knower's knowledge does not help, since, if quantum mechanics be correct, the states of all objects have to be defined in principle by recourse to the concept of probability. Consequently, only if the whole universe is included in the object of scientific knowledge can the qualifying condition "for an isolated system" be satisfied for

even the weaker form of mechanical causation. Clearly, the philosophy of contemporary physics is shown by this book to be as novel in its epistemology as it is in its ontology. Indeed, it is from the originality of its ontology—the consistent unification of potentiality and mechanical causality in its weaker form—that the novelty of its epistemology arises.

Unquestionably, one other thing is clear. An analysis of the specific experimentally verified theories of modern physics with respect to what they say about the object of human knowledge and its relation to the human knower exhibits a very rich and complex ontological and epistemological philosophy which is an essential part of the scientific theory and method itself. Hence, physics is neither epistemologically nor ontologically neutral. Deny any one of the epistemological assumptions of the physicist's theory and there is no scientific method for testing whether what the theory says about the physical object is true, in the sense of being empirically confirmed. Deny any one of the ontological assumptions and there is not enough content in the axiomatically constructed mathematical postulates of the physicist's theory to permit the deduction of the experimental facts which it is introduced to predict, co-ordinate consistently and explain. Hence, to the extent that experimental physicists assure us that their theory of contemporary physics is indirectly and experimentally verified, they *ipso facto* assure us that its rich and complex ontological and epistemological philosophy is verified also.

When such empirically verified philosophy of the true in the natural sciences is identified with the criterion of the good and the just in the humanities and the social sciences, one has natural-law ethics and jurisprudence. In other words, one has a scientifically meaningful cognitive criterion and method for

judging both the verbal, personal and social norms of the positive law and the living ethos embodied in the customs, habits and traditional cultural institutions of the *de facto* peoples and cultures of the world. It is the coming together of this new philosophy of physics with the respective philosophies of culture of mankind that is the major event in today's and tomorrow's world. At this point, the philosophy of physics in this book and its important reference to the social consequences of physics come together.

The chapters of this book have been read as Gifford Lectures at the University of St. Andrews during the winter term 1955–1956. According to the will of their founder the Gifford Lectures should “freely discuss all questions about man's conceptions of God or the Infinite, their origin, nature, and truth, whether he can have any such conceptions, whether God is under any or what limitations and so on.” The lectures of Heisenberg do not attempt to reach these most general and most difficult problems. But they try to go far beyond the limited scope of a special science into the wide field of those general human problems that have been raised by the enormous recent development and the far-reaching practical applications of natural science.